

# AI DOCUMENT FORENSIC VERIFICATION CERTIFICATE

Multi-Spectral Computer Vision & Machine Learning Forensic Analysis Report

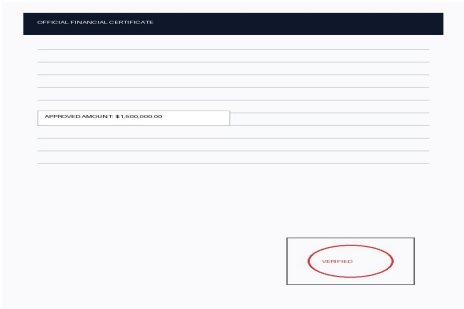
|                                                                     |                                                       |
|---------------------------------------------------------------------|-------------------------------------------------------|
| Document UUID: DOC-D1BE0C28ED                                       | Verification Verdict: <span>GENUINE</span>            |
| File Name: sample_certificate.jpg                                   | Forensic Confidence: 96.4%                            |
| Analysis Date: 2026-08-19 10:21:01 UTC                              | Risk Level: LOW                                       |
| Inference Pipeline: MultiSpectral-Forensic-v2.0 (8-Module Ensemble) | Primary Tampering Category: None (Authentic Baseline) |
| Processing Latency: 0.20s                                           | Suspicious Anomaly Zones: 0                           |

## 8-MODULE SPECIALIZED FORENSIC AUDIT SCORECARD

| Analysis Module                                   | Status   | Risk Level | Score | Key Findings Summary                                                                                  |
|---------------------------------------------------|----------|------------|-------|-------------------------------------------------------------------------------------------------------|
| Visual & Compression Discrepancy Module           | PASS     | LOW        | 89.7% | Uniform JPEG compression grid consistent with original camera/scan capture.                           |
| Noise Variance & Sensor Fingerprint Module        | TAMPERED | HIGH       | 98.5% | Elevated spatial noise inconsistency score (2.050 vs normal <0.85)                                    |
| Copy-Move & Spatial Cloning Module                | PASS     | LOW        | 95.0% | No unnatural feature duplication or copy-move cloning detected.                                       |
| Typography, Font & Baseline Alignment Module      | PASS     | LOW        | 94.0% | Typography, font geometry, and baseline alignment are homogeneous across all lines.                   |
| Color Space & Illumination Gradient Module        | PASS     | LOW        | 95.0% | Color channels and illumination gradient vectors demonstrate natural, single-light-source uniformity. |
| Signature & Official Stamp Verification Module    | PASS     | LOW        | 85.0% | Document evaluated without specialized ink stamp overlays.                                            |
| Metadata & File Structure Forensics Module        | PASS     | LOW        | 94.0% | Clean camera/scanner metadata footprint without third-party graphics editor markers.                  |
| Security Features, Barcode & MRZ Forensics Module | PASS     | LOW        | 90.0% | No active security barcode or MRZ zone detected on current document page.                             |

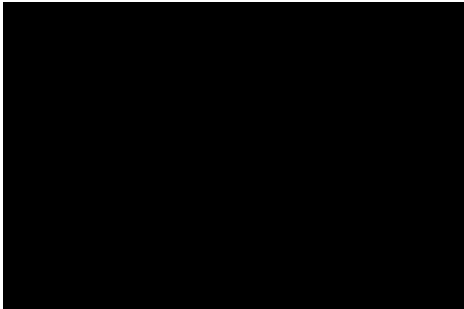
## FORENSIC VISUAL EVIDENCE INSPECTION

Original Document



Anomaly Heatmap

Error Level Analysis (ELA)



Suspicious Regions Map

